🐧 LINUX COMMANDS CHEAT SHEET

NovaShyld Technologies – Network Security Internship

(Task 1, Module 3)

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I. Terminal Navigation

Command	What It Does	Example
pwd	Show current directory	pwd
ls	List files & folders	ls
ls -la	List with details + hidden files	ls -la
cd foldername	Go into a folder	cd /home
cd ..	Go back one level	cd ..
cd ~	Go to home directory	cd ~
mkdir name	Create a new folder	mkdir myfolder
touch file	Create an empty file	touch notes.txt
rm file	Delete a file	rm notes.txt
rm -r folder	Delete a folder	rm -r myfolder
cp file dest	Copy a file	cp notes.txt /tmp
mv file dest	Move or rename a file	mv notes.txt docs.txt
cat file	View file contents	cat notes.txt
clear	Clear the terminal screen	clear

II. File Permissions & Ownership

Command	What It Does	Example
ls -la	View file permissions	ls -la
chmod 644 file	Owner: read+write, Others: read	chmod 644 notes.txt
chmod 755 file	Owner: full, Others: read+execute	chmod 755 script.sh
chmod 777 file	Everyone: full access	chmod 777 folder
chown user file	Change file owner	chown root notes.txt

Command	What It Does	Example
chown user:group file	Change owner and group	chown kali:kali file.txt

Permission Numbers Quick Reference

```
4 = read (r)
2 = write (w)
1 = execute (x)

Examples:
7 = 4+2+1 = rwx (full access)
6 = 4+2   = rw- (read & write)
5 = 4+1   = r-x (read & execute)
4 = 4     = r-- (read only)
```

III. Installing & Updating Packages

Command	What It Does
sudo apt update	Refresh the package list
sudo apt upgrade -y	Upgrade all installed packages
sudo apt install <name> -y	Install a package
sudo apt remove <name> -y	Remove a package
sudo apt search <name>	Search for a package
sudo apt autoremove	Remove unused packages

IV. Basic Networking Commands

Command	What It Does	Example
ifconfig	Show IP addresses & interfaces	ifconfig
ip a	Modern version of ifconfig	ip a
ping -c 4 <IP>	Test connectivity (4 packets)	ping -c 4 8.8.8.8
netstat -tuln	Show open ports & connections	netstat -tuln
traceroute <IP>	Show route to destination	traceroute google.com

Command	What It Does	Example
nslookup <domain>	DNS lookup	nslookup google.com
hostname -I	Show your machine’s IP	hostname -I
wget <url>	Download a file from web	wget http://example.com
curl <url>	Fetch web page content	curl http://example.com